

How Many Firing Rates Should a Neuron Use? Poisson and Non-Poisson Spike Counts in Long and Short Windows

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Abstract—A stimulus sets a neuron’s firing rate anywhere between silence and a peak, and a downstream cell sees only the spike count in a window. The neuron is then a channel from rates to counts, and the rate is its input. The rate code that carries the most information uses finitely many distinct rates; how many was known only between order $\sqrt{A}$ and order A , with A the peak count per window. We conjecture, and test for five to twenty-four rates, that in long windows the number of rates of a Poisson neuron equals the number of levels of the unit-noise Gaussian channel read at the Fisher half-length $\sqrt{A}$, shortened by a constant set at silence. The constant follows from a chain of levels next to silence, with no parameter fitted to the count changes. Certificates in ball arithmetic fix the count at thirty-five peak counts, up to 66 rates at $A = 2000$, well above a $\sqrt{A}$ fit to earlier numerics; four count changes were predicted before they were computed. With a baseline rate the law agrees with twenty-five published count changes with no free parameter, and neurons more regular or more variable than Poisson add one finite-size term. Without an energy budget the long-window code is on-off up to $A = 3.3679$ and graded beyond, and at $A = 10$ an on-off code loses at least 24% of the capacity; under an energy budget at $A = 3$ the code is graded at some budgets and gains at most 0.8%. In short windows the normal approximation, maximized by search, chose an on-off code in all 185 cases computed, many of them outside its accuracy range. At $A = 3$, refractoriness modelled as spike history leaves the capacity at its Poisson value; modelled as count regularity it raises it. A pre-registered test on six datasets did not find discrete levels in recorded neurons.

Index Terms—Capacity-achieving distribution, Poisson channel, neural coding, firing rate, finite blocklength, Fisher information.

I. INTRODUCTION

Consider a neuron whose firing rate a stimulus can set anywhere between silence and a peak rate L . In a counting window of length T the neuron emits a random number of spikes with mean xT , where x is its current rate, and a downstream reader sees only that count. The rate is therefore the neuron’s input and the count is its output, so the neuron is a noisy channel from rates to counts. We write $A = LT$ for the peak count, the largest expected number of spikes in one window.

When spikes are counted over many windows, the distribution of rates that carries the most information about the stimulus uses finitely many distinct rates, one of which is

silence [1], [2]. An optimal rate code thus drives the neuron at a few discrete levels, and theory has predicted discrete tuning curves, from the information they carry [4] and from the error of reconstructing the stimulus [3]. The levels matter for energy as well, because every spike costs ATP [5] and the level structure is what a metabolic budget buys [6]. How many levels the optimal code uses has been known only within wide bounds, between order $\sqrt{A}$ and order A for a Poisson neuron [7], [8].

Three features of a real neuron could change that number. Its spike count may be more regular than Poisson, because of refractoriness, or more variable; it may fire at a baseline rate even when the stimulus asks for silence; and the reader may count spikes over a window that holds only a few of them, where the asymptotic notion of capacity no longer applies. This paper treats all three, in the terms of Table I; a *count change* is the peak count at which the optimal code gains a rate.

- In long windows, we conjecture that the number of rates of the optimal code equals the number of levels of the Gaussian channel with unit noise, read at the half-length $\sqrt{A}$ shortened by a constant set at silence (Section III). The constant is computed from a chain of levels next to silence, with no parameter fitted to the count changes, and the conjecture is tested for five to twenty-four rates. Certificates in ball arithmetic fix the exact count at thirty-five peak counts, up to 66 rates at $A = 2000$, and four count changes were predicted before they were computed. The certified counts at large A lie well above a $\sqrt{A}$ fit to earlier numerics, which had suggested $\sqrt{A}$ growth.
- Neurons more regular or more variable than Poisson carry the same constant at silence, and their counts follow the same law with one finite-size term (Section IV).
- A baseline rate moves the constant from its Poisson value to the Gaussian one; with no free parameter, the law reproduces twenty-five published count changes (Section V).
- Without an energy budget the long-window code is on-off up to the first count change, $A = 3.3679$, and graded beyond it, where the graded code soon carries much more: at $A = 10$ an on-off code loses at least 24% of the capacity. Under an energy budget at $A = 3$ the code is graded at some budgets and gains at most 0.8% there. For

short windows we maximized the normal approximation to the finite-blocklength rate; for the Poisson neuron and for neurons with a refractory spike history its maximizer was binary in all 185 cases computed, at peak counts up to 6, many of them outside the accuracy range of the approximation (Section VI).

- Refractoriness modelled as spike history cannot change the probability of a silent window, so at $A = 3$, where the optimal codes are on-off, such a neuron has exactly the Poisson capacity; on the energy–capacity front it lies above the Poisson neuron and below a count model of similar regularity (Section VI).
- A pre-registered test on six public datasets did not find the predicted levels in recorded neurons (Section VII).

Most of these statements are measurements. The certified counts are theorems about single peak counts, and one interval statement follows from them under a published monotonicity conjecture (Section III); the count law itself is a conjecture supported by the evidence below. Predictions marked as registered were written in a dated, append-only log before the quantity was computed; the order is asserted by us from file times. Table IV grades every claim. The scripts, the certificates and the result files are at <https://github.com/Tristan-tech-ai/how-many-firing-rates>, and the log is public with the companion paper’s material [18].

What the answer is for, and what it does not show. For neuroscience the law turns the theory of optimal rate codes into numbers that can be checked against a neuron. For a neuron with a peak rate of 100 spikes/s read in windows of 30 ms ($A = 3$), the information-maximizing code is on-off; in windows of 100 ms ($A = 10$) it uses four rates, since the change from three to four rates lies near $A = 9.8$ in the computed optima of Barletta and Dytso [17] and a fifth rate first appears at $A = 18.1$ (Table II). There the graded code matters: an on-off code carries at most $\log 2$ nats per window, at least 24% less than the capacity at $A = 10$. At $A = 3$ under an energy budget, where a graded code can also win, it gains less than 1%. In short windows the objective we maximized chose an on-off code in every case, but most of those cases lie where that objective is inaccurate (Section VI). The same channel describes photon counting with a peak-limited light source and dark counts [1], where the baseline is the dark-count rate: the conjectured law predicts the number of intensity levels of the best pulse-amplitude constellation without solving for it, and it agrees with published dark-current results [17]. The test on recorded neurons is negative. Either these neurons do not maximize this information at the windows tested, or the data had too few trials to show the levels; we cannot tell which. The law concerns one neuron whose windows are independent uses of the channel. It says nothing about populations, about codes that use spike timing, or about what a downstream cell actually reads.

II. THE NEURON AS A CHANNEL

The input $x \in [0, A]$ is the expected count produced by the stimulus in one window, the output Y is the count, and $Y \sim \text{Poisson}(x + \lambda)$, where $\lambda \geq 0$ is the baseline count per window

(the dark current of the optics literature). For an input law P with output mixture Q , let $D(x) = D(\text{Poisson}(x + \lambda) \| Q)$. Then P achieves the capacity C if and only if

$$D(x) \leq C \text{ on } [0, A], \text{ with equality on } \text{supp } P, \quad (1)$$

and the optimal P is unique and has finitely many atoms [1], [9]. We write $N(A, \lambda)$ for their number and $N(A) = N(A, 0)$. The Fisher information of the Poisson count is $1/(x + \lambda)$, so in the Fisher coordinate $t = 2\sqrt{x + \lambda}$ the noise has unit size everywhere: two rates one unit apart in t are equally hard to tell apart wherever they lie. The input range has half-length

$$h(A, \lambda) = \sqrt{A + \lambda} - \sqrt{\lambda}. \quad (2)$$

The reference is the Gaussian channel $Y = X + Z$ with $Z \sim N(0, 1)$ and $X \in [-A_g, A_g]$, whose input is discrete at the optimum [24], whose Fisher coordinate is its input itself and whose half-length is A_g . Read in the Fisher coordinate, the Poisson channel approaches this Gaussian channel at large counts [25], and Jeffreys’ prior, uniform in that coordinate, is asymptotically capacity-achieving [26]. We write $N_G(h)$ for its number of levels at half-length h , a staircase in h , and $A_g(K)$ for the half-length at which its K -th level appears.

Two families describe neurons that are not Poisson. A count more regular than Poisson has Fano factor below one; the binomial count is the simplest case, with Fano factor $1 - p$. A count more variable than Poisson has Fano factor above one, the negative binomial being the standard case, with Fano factor $1 + \text{mean}/\rho$ for size parameter ρ . The Conway–Maxwell–Poisson (COM-Poisson) family [11] covers both directions with one shape parameter ν , sub-Poisson for $\nu > 1$ and super-Poisson for $\nu < 1$, and contains the Poisson count at $\nu = 1$; it is a standard model of spike counts [12]. Refractoriness can also be modelled as spike history, by a generalized linear model (GLM) whose conditional intensity is multiplied by $\exp(\sum_j \eta_j y_{t-j})$, where y_{t-j} marks a spike j bins earlier and $\eta(s) = -a e^{-s/\tau}$ at lag s . We use $\tau = 4$ ms, amplitudes $a = 0, 2, 4$ and 8 , and bins of 2 ms, and the history is reset at the start of each window. A bin holds at most one spike, so even the kernel $a = 0$ gives a count with Fano factor 0.89 at a rate of 60 Hz in 50 ms windows; the GLM without history is therefore not the Poisson neuron.

In a short decision the reader does not see many windows. If a message must be decoded from n windows (the blocklength) with error probability ϵ , the largest number of messages M^* satisfies, in the normal approximation that drops terms of order $\log n$,

$$\log M^*(n, \epsilon) \approx nC - \sqrt{nV} Q^{-1}(\epsilon), \quad (3)$$

where V is the channel dispersion, the variance of the information density, and Q^{-1} the inverse of the Gaussian tail function [13]; with an energy cost the same form holds with the cost-constrained C and V [14]. We charge energy linearly in the mean rate, as the Attnell–Laughlin budget does [5]. Table I collects the symbols.

III. LONG WINDOWS: HOW MANY RATES

A. The count law

In the Fisher coordinate the levels of the optimal Poisson code are spaced like the levels of the unit-noise Gaussian

TABLE I
SYMBOLS

symbol	meaning
$x, A = LT$	expected count in a window; peak count
λ	baseline count per window
$N(A, \lambda), N_G(h)$	number of rates (Poisson); of levels (Gaussian)
$h = \sqrt{A + \lambda} - \sqrt{\lambda}$	Fisher half-length of the rate range
$A_K, A_g(K)$	peak count (Poisson) and half-length (Gaussian) at which level K appears
$c(K)$	wall constant, $\sqrt{A_K} - A_g(K)$
n, ϵ, V	number of windows, error probability, dispersion

channel, except near silence. A silent neuron never spikes, so the output at $x = 0$ is degenerate, and it sits at a finite Fisher distance from the rest of the range. In the computed optima the levels next to silence form a boundary layer, and the law below treats its effect as an extra length that changes slowly with K . For the Gaussian channel Zhang argued that the optimal input approaches Jeffreys' prior in the bulk, with a discrete pattern next to the boundary [42] (as reported in [30]), and Abbott and Machta found the first levels next to an endpoint at fixed Fisher distances [29]. We conjecture that the Poisson count is the Gaussian count read at a shortened half-length,

$$N(A, \lambda) = N_G(h(A, \lambda) - c(K, \lambda)), \quad (4)$$

where $c(K)$ is the wall constant. Since K is the count being predicted, (4) is implicit, and we evaluate it at the count changes A_K , where K is known. We call silence a *wall* because the constraint $x \geq 0$ is active there: the level at silence carries no derivative condition in (1). The constant is defined from the measured count changes as $c(K) = \sqrt{A_K} - A_g(K)$. It can also be predicted without them, from a half-line chain: one level fixed at silence with a free weight, thirty free levels, and ten frozen levels of unit weight at spacing g in the Fisher coordinate, 41 levels in all, more than the full problem has for $K \leq 24$. The chain's equations ask that D take the same value at every free level and at the first frozen one, and that $D' = 0$ at the free levels: 61 equations for the 30 positions and 31 weights. We solve them once with the Poisson kernel and once with the Gaussian kernel, and define the *excess* $E(g)$ as the Fisher position of the thirtieth free level in the Poisson chain minus its position in the Gaussian chain. We call the resulting prediction the *closure*:

$$\begin{aligned} c(K, \lambda) &= \frac{1}{2} E(g_{\text{eff}}(K), \lambda), \\ g_{\text{eff}}(K) &= A_g(K + 1) - A_g(K - 1), \end{aligned} \quad (5)$$

where the factor $\frac{1}{2}$ converts the Fisher coordinate $t = 2\sqrt{x}$ to the scale of the half-length $\sqrt{A}$, and g_{eff} is the local spacing of the Gaussian levels at the count change. The chain law $E(g) = 0.9840 - 0.2347g$ is a fit to the chain's own excess on $1.43 \leq g \leq 2.35$ (root-mean-square deviation 0.0012); it never sees a count change. The estimator g_{eff} was chosen among three by comparing them with the measured wall constants; the other two, the one-sided spacing $2(A_g(K + 1) - A_g(K))$ and the mean spacing $2A_g(K)/(K - 1)$, deviate by 0.0045 and 0.0312 against 0.0017.

TABLE II
COUNT CHANGES OF THE POISSON NEURON AND THE WALL CONSTANT.
 A_g : GAUSSIAN HALF-LENGTH; A_K : POISSON PEAK COUNT;
 $c = \sqrt{A_K} - A_g$; CLOSURE $E(g_{\text{eff}})/2$. ROWS 21–24 WERE PREDICTED
BEFORE THEY WERE COMPUTED; THE CLOSURE COLUMN OMMITS THE
+0.002 CORRECTION USED FOR THE PREDICTIONS OF ROWS 23–24.
 $A_g(15)$ TO $A_g(26)$ ARE VALUES RELOCATED AT 128 BITS.

K	$A_g(K)$	A_K	$c(K)$	closure	difference
5	4.02128	18.1366	0.2374	0.2397	−0.0022
6	5.05816	28.2020	0.2524	0.2551	−0.0027
7	6.03983	39.7395	0.2641	0.2666	−0.0025
8	6.97912	52.5990	0.2734	0.2756	−0.0023
9	7.88348	66.6669	0.2815	0.2831	−0.0016
10	8.75930	81.8518	0.2879	0.2894	−0.0015
11	9.60970	98.1411	0.2969	0.2949	+0.0021
12	10.43922	115.3029	0.2987	0.2997	−0.0010
13	11.24861	133.5236	0.3066	0.3039	+0.0027
14	12.04221	152.5030	0.3070	0.3078	−0.0008
15	12.81867	172.4022	0.3115	0.3113	+0.0002
16	13.58195	193.0849	0.3136	0.3144	−0.0008
17	14.33238	214.6284	0.3178	0.3173	+0.0006
18	15.07098	236.8398	0.3186	0.3199	−0.0013
19	15.79863	259.9253	0.3236	0.3224	+0.0012
20	16.51609	283.6563	0.3260	0.3247	+0.0013
21	17.22406	308.0874	0.3284	0.3269	+0.0015
22	17.92312	333.1959	0.3305	0.3289	+0.0016
23	18.61381	358.9655	0.3326	0.3308	+0.0018
24	19.29662	385.3810	0.3345	0.3326	+0.0019

A worked case shows how the law is used. For $K = 15$, Table II gives $A_g(14) = 12.0422$ and $A_g(16) = 13.5819$, so $g_{\text{eff}} = 1.5397$, $E = 0.6226$ and $c = 0.3113$. Adding c to $A_g(15) = 12.8187$ gives $\sqrt{A_{15}} = 13.1300$, against the measured 13.1302: the fifteenth rate of the optimal code appears at a peak count of 172.40. Nothing in that line was fitted to a Poisson count change.

B. The evidence

Table II lists the count changes for $K = 5$ to 24. The wall constant rises from 0.237 to 0.335 with steps that shrink, and the closure (5) matches it with a root-mean-square deviation of 0.0017, over $K = 5$ to 20 and over $K = 5$ to 24 alike. The residual is systematic at that level: negative for every K from 5 to 10, positive for $K = 19$ to 24.

Four count changes were registered before they were computed, each with a tolerance of ± 0.15 in A . The twenty-first was registered at $A = 307.99$ from an earlier local fit of the closure; the full-range law above had been recorded before the computation, and it gives 308.03, a number first written down after it. The twenty-second was registered at 333.14 by the full-range law. The computation returned 308.09 and 333.20. Both residuals, +0.0015 and +0.0017 in c , matched the size of a finite-length correction, +0.002, that had been estimated beforehand from chains of twenty, thirty and forty free levels, and only then did we adopt it. With the correction the law placed the next two at 358.97 and 385.39, and without it at 358.90 and 385.31; both sets were registered, and the computation returned 358.97 and 385.38 (Fig. 1). The law has not been evaluated at the certified 46 and 66 rates of Section III-C.

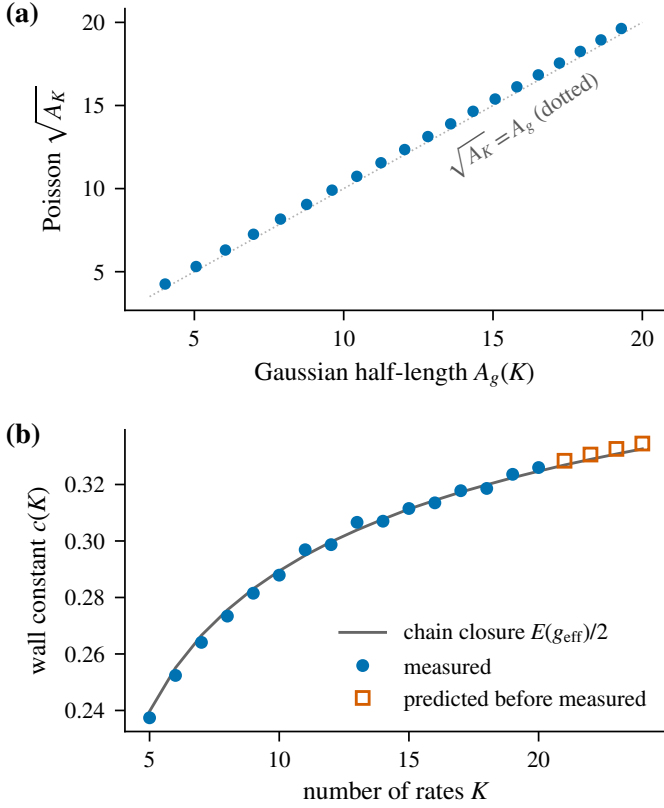


Fig. 1. The Poisson count is the Gaussian count at a shortened Fisher half-length. (a) Poisson count changes $\sqrt{A_K}$ against the Gaussian ones $A_g(K)$, $K = 5$ to 24 : a line of slope one, offset by the wall constant. (b) The wall constant $c(K)$, measured (dots) and predicted by the chain closure (line); open squares, $K = 21$ to 24 , were predicted before they were computed.

C. Certified counts

The count itself was certified at thirty-five peak counts. For each, the optimality condition (1) was proved over the whole range $[0, A]$ in ball arithmetic for one computed input [15], and since the optimal input is unique this fixes $N(A)$ exactly. Thirty-two certificates cover $A = 27.9$ to 378 with five to twenty-three rates, none at nine (Appendix A); among them the smallest margin by which D stays below C off the support is 3.3×10^{-13} , at $K = 22$. Three later certificates give 21 rates at $A = 311.37$, 46 at $A = 1144.25$ and 66 at $A = 2000$, with margins 1.9×10^{-20} , 1.8×10^{-25} and 1.8×10^{-29} . These are theorems about single peak counts. One statement about intervals follows if the count never falls as A grows, which is half of a conjecture that Sharma and Shamai made from numerical evidence for related peak-limited channels [16].

Theorem 1 (conditional on monotonicity). *If $N(A)$ is non-decreasing, then $N(A) = K$ on each of the eleven intervals between the smallest and largest certified peak counts with the same K in Table V; together they cover 170.4 of the 350.1 in A spanned by these certificates.*

Proof. If $N(a) = N(b) = K$ with $a < b$, then $K = N(a) \leq N(A) \leq N(b) = K$ for every $A \in [a, b]$. $\square$

The later certificate of 21 rates at $A = 311.37$ lies inside the interval for $K = 21$, as the theorem requires. The published

staircases of Barletta and Dytso [17] show no fall in 6430 consecutive pairs at four baselines, 1320 of them without a baseline. These are numerical optima on a grid of step 0.1 in A , up to $A = 127.5$ without a baseline, so a narrower dip would not be seen. Across the 230 consecutive pairs of high-precision states on the nine plateaux we sampled, neither the smallest gap between rates nor the smallest weight ever decreases. A fall would need one of them to reach zero, so this argues against a fall, but it is not a proof.

D. How fast the count grows

The law (4) carries the growth of the Gaussian count to the Poisson neuron. Abbott and Machta derive, from a continuum ansatz and without proof, that the Gaussian count grows as $L^{4/3}$ at half-length L [28], [29], so through the Fisher length the Poisson count would grow as $A^{2/3}$; the best proved bounds on the Gaussian count are of order $L\sqrt{\log L}$ from below and L^2 from above [30], [31]. In the companion paper the leading constant is measured, as 0.4002, in three problems [18], so the Poisson neuron should use about $0.4002 A^{2/3}$ rates at large A . This gives 18.4, 43.8 and 63.5 at the three later certificates, against the certified 21, 46 and 66; the Gaussian count exceeds its own leading term by a similar amount, three levels at $L = 12.8$ and 19.3 . From numerical optima up to $A = 128$, Barletta and Dytso conjectured that the Poisson count grows as $\sqrt{A}$ [17], the order of a lower bound that Dytso, Barletta and Shamai had conjectured to be tight [7]. A $\sqrt{A}$ law fitted to those optima for $A \geq 20$ gives 37 and 49 rates at $A = 1144.25$ and 2000 , against the certified 46 and 66, while an $A^{2/3}$ law fitted to the same optima gives 47 and 68. No finite range settles an asymptotic order, but over this range the counts follow $A^{2/3}$, not $\sqrt{A}$. For every $c < 0.0117$, a proof that the count is at least $cA^{2/3}$ for all A beyond a threshold, which is not computed and is very large, has been written but not yet independently reviewed.

At the other end, the count changes first from two rates to three at $A = 3.3679$, as Shamai showed [1] (as quoted in [17]): below it the best code is on-off, silence and the peak rate. For mutual information, Bethge et al. noted without showing the computation that a binary code stays optimal up to a peak count of about 3.5 [3], and conditions for binary inputs with a baseline or an average-power constraint are in [10]. We recomputed 3.3679 from the interior optimality condition of the best binary input. Section VI returns to it.

IV. NEURONS MORE REGULAR OR MORE VARIABLE THAN POISSON

A neuron more regular or more variable than Poisson keeps the wall at silence. The binomial and negative-binomial counts both have a degenerate output at silence, at a finite Fisher distance, and the first two gaps next to silence, the spacings between neighboring rates in the Fisher coordinate [27], are the Poisson ones: 3.126 and 2.260 Fisher units for the binomial with $n_b \geq 1000$ trials, and 3.125 and 2.260 for the negative binomial with size $\rho = 1000$. These place the first two levels above silence at 3.13 and 5.39; Abbott and Machta found 3.13 and 5.41 for the Poisson count [29]. Unlike the Poisson count,

whose variance equals its mean, these counts have a variance-to-mean ratio that changes with the rate, and that adds one term. Writing $q(x) = \sqrt{\text{mean}/\text{variance}}$ and $q_0 = q(0)$, the count is read at the effective half-length

$$h_{\text{eff}} = h_F - \kappa \frac{h_F - q_0 \sqrt{A}}{K}, \quad (6)$$

where h_F is the Fisher half-length, A the peak count and $\kappa = 0.42 \pm 0.02$ a measured constant: 56 count changes of the binomial count on $[0, \varrho n_b]$, a fraction ϱ of its full range with peak count $A = \varrho n_b$ and one wall ($K = 9, 11, 17$ and 18), nine of the binomial channel on its full range, where both ends are walls, and six of the negative binomial, which gives 0.39. The term vanishes for the Poisson count and for any count whose variance-to-mean ratio is constant, as the generalized Poisson count of Consul confirms ($|\Delta h| < 0.0005$), and it vanishes as $K \rightarrow \infty$, so the Fisher-length law is its large- K limit. Registered before they were computed, (6) placed the 25th and 26th count changes of the full-range binomial channel at $n_b = 174.94$ (with $\kappa = 0.435$, the value measured at the time) and 186.36 (with $\kappa = 0.42$); $\kappa = 0.42 \pm 0.02$ allows 174.76–174.97 and 186.24–186.45, and the solver returned 174.934 and 186.386. At $\kappa = 0.42$ the two predictions are low by 0.07 and 0.04. For the binomial channel, the published bounds on the count and the computed optimal inputs are in [32]–[34].

The COM-Poisson neuron also has a degenerate output at silence, so the same construction predicts its count, with its own Fisher length and dispersion ratio; we have not computed its wall constant. What we did compute is its count at the Fano factors measured in the datasets of Section VII, 0.90 to 1.13: for $A \leq 12$ it differs from the Poisson count by at most one rate, and variability above Poisson lowers it. In long windows, then, the number of rates is a property of the Fisher length of the rate range and of the type of the endpoint at silence, and a neuron's Fano factor enters only through these and, at finite K , through the dispersion ratio at silence in (6).

V. A BASELINE RATE

A baseline rate removes the degenerate output: with $\lambda > 0$ a silent stimulus still produces spikes, and the endpoint at silence becomes regular. The wall constant then falls from its Poisson value toward the Gaussian value, zero; at $K = 11$ it is 0.30 at $\lambda = 0$ and 0.03 at $\lambda = 1$. The chain with the same kernel predicts this crossover with no parameter, to 0.002 at $K = 11$ (six baselines) and to 0.001 at $K = 15$ (three). In the Fisher offset of the endpoint, $d = 2\sqrt{\lambda}$, the change is linear at small d , $E(g, 0) - E(g, \lambda) = 1.06d$, and the remaining excess decays as $0.055/d$ at large d ; a baseline of 10^{-3} spikes per window already removes a tenth of the wall.

The law (4)–(5) with this chain predicts the twenty-five count changes in the published optimal inputs of Barletta and Dytso [17] for $\lambda = 1, 10$ and 100 (nine, eight and eight changes, from three rates up to eleven). The root-mean-square deviations are 0.005, 0.004 and 0.003 in the Fisher half-length, and the largest is 0.009. The six changes to three and four rates lie below $K = 5$, where g_{eff} exceeds the range on which the chain law was fitted; for them the agreement is with the law's extrapolation. The chain also gives the second rate of the

code: its first gap g_1 next to silence places it at an expected count of $(g_1/2 + \sqrt{\lambda})^2 - \lambda$ above the baseline. This is 4.64, 10.48 and 29.05, against the published 4.64, 10.49 and 29.06 at the largest peak count of each run. For every baseline the lowest rate of the optimal code stays at silence, where the neuron fires at the baseline rate. Where a new rate appears is also set at silence: an inserted rate appears at 0.62 of its gap from the silent side, and a split happens at 0.12 of the gap next to the parent rate, against $\frac{1}{2}$ and 0 for the Gaussian channel; the baseline carries both values to the Gaussian ones (Appendix B).

VI. SHORT WINDOWS

A. A window that holds few spikes

Spike counts are commonly computed in windows of about 50 ms [35], and such a window often holds only a few spikes. There the asymptotic capacity overstates what a neuron delivers; Kostal and Shinomoto found a drop of about an order of magnitude under finite coding [21]. By (3), the relative loss $\phi = \sqrt{nV} Q^{-1}(\epsilon)/(nC)$ of a Poisson neuron depends mainly on the expected number of spikes N_{sp} in the whole decision, $\phi \approx Q^{-1}(\epsilon)/\sqrt{N_{\text{sp}}}$ [19]. The approximation holds when C^2/V is close to the mean count per window, since $\phi = Q^{-1}(\epsilon)\sqrt{V/(nC^2)}$. In the windows used here C^2/V is about 12% above the mean count (at $A = 3$, $C = 0.5944$ and $V = 0.2224$ nats, so $C^2/V = 1.59$ against a mean count of 1.41). In continuous time with a slack power constraint the two are equal [36]; when an energy budget binds they are not, and the approximation fails. At $\epsilon = 0.01$, where $Q^{-1}(\epsilon) = 2.33$, a decision built on fifty spikes loses a third of the asymptotic rate. With fewer than about sixty spikes ϕ exceeds 0.3, where the expansion is no longer accurate and overstates the loss, but the loss stays large. Below a total of about $Q^{-1}(\epsilon)^2 = 5.4$ spikes the right side of (3) is negative, that is, less than one message can be decoded reliably, a known consequence of the expansion.

B. The best short-window code is on-off

The loss also changes which code is best. Bethge, Rotermund and Pawelzik found binary tuning optimal for rate coding at high temporal resolution [3]; here the objective is the right side of (3), and the comparison is with the exact long-window optimum. We maximized that objective by search over input laws on a grid of rates from 0 to 60 Hz, for the Poisson neuron and for the four GLM kernels of Section II (induced Fano factors 0.89 to 0.46): 130 cases over energy budgets and blocklengths $n = 10$ and 50 at $A = 3$, in windows of 50 ms, and 55 at a fixed decision time of 500 ms, with windows of 10 to 100 ms, that is, peak counts from 0.6 to 6 and n from 50 down to 5, eleven of them for the Poisson neuron. In all 185 cases the maximizer was binary: silence and one firing rate. A graded law was open to the search in every case and was never chosen. Three limits apply. The maximizer was found by search and is not certified. The cases at $n = 10$ and all 55 at fixed decision time hold at most 30 expected spikes per decision, so $\phi \approx Q^{-1}(\epsilon)/\sqrt{N_{\text{sp}}}$ is at least 0.4 there, outside the accuracy range of (3). And in many cases

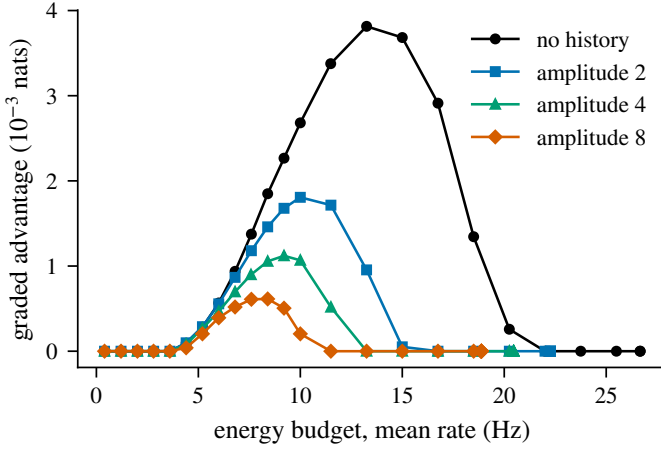


Fig. 2. Under an energy budget the long-window code can be graded below the first count change, but it gains little. Peak count $A = 3$ (60 Hz, 50 ms windows): advantage of the Blahut–Arimoto optimum over the best binary code on a 0.5 Hz grid, for a neuron without spike history and refractory amplitudes $a = 2, 4, 8$ ($\tau = 4$ ms); zero means that no graded code was shown. The maximizer of the normal approximation was binary in every case.

the long-window code is on-off as well, so they cannot show a contrast: at fixed decision time only the peak counts 3.5, 4.5 and 6 exceed the first count change of the Poisson neuron. An exact finite-blocklength bound for a few cases would show whether the on-off choice survives outside the approximation.

The long-window optimum behaves differently. Without an energy budget it is binary up to $A = 3.3679$ and graded beyond, so it changes at the first count change of Section III, and beyond it the graded code soon carries much more than any on-off code. An on-off code carries at most $\log 2 = 0.693$ nats per window, while the capacity is at least 0.762 nats at $A = 6$, 0.916 at $A = 10$ and 1.106 at $A = 18$ (Blahut–Arimoto on 601 rates), so an on-off code loses at least 9%, 24% and 37% of the capacity there; at $A = 4$ the best on-off code loses 1.1%. With an energy budget the code can be graded at smaller peak counts. At $A = 3$ we compared the Blahut–Arimoto optimum with every binary code on a 0.5 Hz grid and call a budget graded when the former exceeds the latter. For the neuron without spike history the long-window optimum is graded at budgets from 4.4 to 20.2 Hz (Fig. 2). Over the four kernels, 42 of 83 distinct budgets are graded in this sense; at the others a graded optimum is not shown, but not excluded either, since the Blahut–Arimoto value there falls up to 1.5×10^{-4} nats below the best binary code. Where a graded code was found it gains at most 3.8×10^{-3} nats per window over the best binary code, 0.8% of the cost-constrained capacity at that budget. Near $A = 3$ the objective is nearly flat between graded and on-off codes, so a small dispersion term is enough to tip the choice to on-off. A prediction that rests on a nearly flat objective is fragile, and at $A = 3$ we report the effect as a change in which code is optimal; the information at stake there is small.

For a physiologist the switch is a statement about two numbers, the peak rate and the counting window. Without an energy budget the long-window code is graded only when $LT > 3.3679$, which at a 50 ms window needs a peak rate of

67 Hz. Driven layer-4 neurons of awake trained animals fire at a mean of 12 Hz, and at 28.5 Hz one standard deviation above that mean [20], below that, while high-rate cells or long counting windows can pass it. Without an energy budget, a neuron that cannot pass it uses the on-off code in long windows, and the short-window objective chose on-off as well.

C. Refractoriness as spike history and as count regularity

A refractory kernel acts only after a spike. When the history is reset at the start of each window, the probability of a silent window is therefore e^{-xT} for every kernel, the void probability of a process with an empty history [37]; at $xT = 3$ we computed $0.04978707 = e^{-3}$ for all four kernels. With an on-off code any count above zero identifies the active rate, so the information depends on the kernel only through that probability. At $A = 3$, where the capacity-achieving code of each of the four GLM channels is on-off (computed), every GLM neuron therefore has exactly the Poisson capacity, 0.5944 nats per window. Above the first count change the Poisson optimum is graded and the identity says nothing. For their continuous-time channel, Kostal and Shinomoto conclude more generally that a non-Poissonian neuron cannot increase the capacity through causal feedback, provided that the input satisfies their peak and average constraints and that the spontaneous activity is Poissonian [21]; the silent-window identity is the mechanism in our setting. A receiver-side dead time behaves differently and does lower the capacity [38], and the capacity per unit cost of a refractory counting model is studied in [39]. Spike timing inside the window adds at most 2.7×10^{-4} nats per window over the count, so the count is nearly sufficient.

Refractoriness does lower the energy cost, because it suppresses spikes: the realised rate of the top symbol falls from 56.5 to 40.0 Hz as the kernel strengthens. At a matched mean rate between 3 and 25 Hz, in long windows, the GLM neuron with induced Fano factor 0.46 therefore beats the Poisson neuron by 0.034 to 0.072 nats per window. A COM-Poisson count of similar Fano factor (0.5) does better again, by 0.026 to 0.118 nats per window over the GLM neuron, because it also lowers the probability of a silent window, which the GLM leaves at its Poisson value: at Fano factor 0.5 and a mean count of 2.83 that probability is 0.0089, against 0.059 for a Poisson count of the same mean. Refractoriness as spike history and as count regularity therefore behave differently, and a count model of a refractory neuron overstates what refractoriness buys (Fig. 3(b)).

D. The energy–capacity front

The energy–capacity front plots the best information per window against the mean firing rate it costs; Kostal and Shinomoto drew it for the Poisson neuron on the Atwell–Laughlin axis [21], and energy–information trade-offs of neurons go back further [6], [40], [41]. For long windows the front is the capacity–cost function, computed by cost-constrained Blahut–Arimoto, so each point is a global optimum. For short windows the front was found by search over input laws under a direct constraint on the mean rate. A Lagrangian sweep cannot

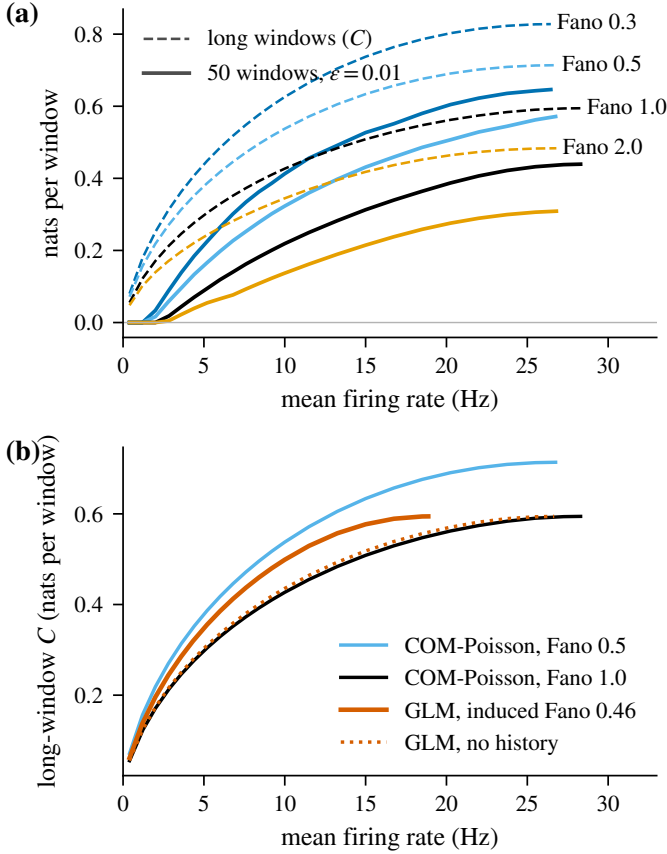


Fig. 3. The energy–capacity front of a neuron. (a) COM-Poisson neurons with Fano factors 0.3 to 2: long-window capacity (dashed) and the short-window front with 50 windows at $\epsilon = 0.01$ (solid, $\log M^*$ per window). Lower Fano factor is better at every energy but one (43 of 44). (b) Long-window fronts of COM-Poisson neurons and of GLM neurons: refractory spike history lies above the Poisson neuron but below a count model of similar regularity. Windows of 50 ms, rates 0 to 60 Hz.

replace the direct constraint here: the short-window objective is not concave in the input, and at $n = 10$ a sweep over 121 multipliers never reaches rates between 5.32 and 22.34 Hz. Fig. 3(a) shows both fronts for COM-Poisson neurons with Fano factors from 0.3 to 2; both rise smoothly with the energy budget. Lower Fano factor is better at every energy but one in both (43 of 44 comparable energies; the exception, Fano factors 1.5 and 2 at $n = 50$ and 3.6 Hz, survived a tenfold finer search), so the dispersion term does not reorder the neurons. The efficient operating point moves, however. On the Attwell–Laughlin budget, which adds a static cost to the cost per spike, the information per ATP of the Poisson neuron in long windows is largest at 1.2 Hz among the three budgets computed (0.4, 1.2 and 2 Hz), near the mean rate of about 4 Hz that the Attwell–Laughlin budget assumes [5], but it is largest near 10 Hz with 50 windows. Short decisions thus push the efficient rate up, away from the measured rates; for a single neuron the binding variable is the number of spikes in the decision, which points to pooling or to longer integration.

E. What Kostal and Shinomoto found, and what changes here

Kostal and Shinomoto computed the energy–capacity trade-off of a Poisson neuron with the Attwell–Laughlin cost,

together with a version for finite coding at fixed error probability [21]. They also asked whether, under some coding constraint, a continuously varied rate might help; the graded long-window codes above are one answer, with a small gain at $A = 3$ and a large one beyond the first count change. Three differences in the setup explain the differences in the results. First, their input may switch freely in continuous time, which makes the on-off code optimal; we hold the rate fixed through a window, and only this restriction allows a graded code at all. Second, they cap the peak rate at 50 Hz to keep the linear cost model and the Poisson approximation valid. At a 50 ms window that cap is $A = 2.5$, below 3.3679, so without an energy budget our long-window code is on-off there as well; their model carries an average-power constraint besides the peak, their Eq. (2), and with a budget our code can be graded at $A = 3$ (Fig. 2), so whether it is graded at $A = 2.5$ under their budget is not settled here. Third, their finite-coding version comes from an error exponent: the window grows until a target error is reached, a bandwidth-type constraint under which a low rate is paid for with time, and their efficient operating point moves to lower cost. Ours fixes the number of windows, a blocklength-type constraint that penalizes low rates instead, and the efficient point moves to higher rates. The two constraints bound the operating point from opposite sides. Our neuron family is also wider, since the COM-Poisson count covers both directions of variability and contains their Poisson neuron as one member.

VII. DO RECORDED NEURONS USE DISCRETE LEVELS?

Two lines of theory predict that a single neuron’s tuning curve takes a few discrete values [3], [4]. Nikitin et al. note that it is not clear whether a test with the right stimulus and counting window has been done, and Bethge et al. cite evidence for binary coding in some neurons, such as the H1 neuron of the fly. We ran a pre-registered test on six public datasets: motor cortex, area 2 and dorsomedial frontal cortex of monkeys (DANDI 000128, 000127, 000130 [22]), larval salamander retina (Dryad 10.5061/dryad.4qrfj6qm8), mouse V1 (Allen Visual Coding Neuropixels, DANDI 000021) and mouse A1 (DANDI 000986). For each tuned unit the tuning map was estimated from the trials, and the number of levels was the number of k-means clusters of that map that best predicts held-out trials. If the code were discrete, this number would stop growing once enough trials were added. Two comparators were built on each unit: a discrete one, with $N(A, \lambda)$ levels on the unit’s own map and matched noise, and a continuous one, with the map itself and the same noise. Only units whose two simulations separate are scored, and a shuffle control must give one level.

The registered rule returned *mixed* in five datasets and *smooth* in the salamander retina (Table III). Every *mixed* verdict came from units without tuning, which the rule’s tolerance, fixed in advance, counted as flat, so we also report a second reading restricted to tuned units. There the level count grew with the number of trials in 57 to 85% of the tuned units of five datasets; in mouse A1, where the count is capped at five levels, 70% of tuned units reached all five. At the full

TABLE III
THE LEVEL-COUNT TEST ON SIX PUBLIC DATASETS

dataset	rule	tuned	levels (data / disc. / cont.)	growth	flat at 2
M1 (MC_Maze)	mixed	136	8.9 / 2.7 / 8.7	79%	1–4%
area 2	mixed	65	8.1 / 3.0 / 6.6	57%	4%
DMFC	mixed	53	5.4 / 3.0 / 7.4	59%	6–13%
salamander retina	smooth	381	6.6 / 3.1 / 8.4	75%	4–6%
mouse V1	mixed	100	6.5 / 2.7 / 7.9	85%	8%
mouse A1 (ceiling 5)	mixed	178	4.3 / 2.3 / 4.1	70% ^a	8%

rule: verdict of the registered rule; other columns: tuned units only.
^aunits reaching all five levels.

trial count the level count was 1.8 to 3.3 times the discrete comparator; it was within 0.2 of the continuous comparator in two datasets and differed from it by 1 to 2 levels in the other four, in both directions, so the comparators calibrate the test only to about two levels. Only 1 to 13% of tuned units stopped at two levels, the number the theory predicts at the smallest peak counts. The A1 recordings, with a peak count of about 0.37 per window and about 1490 repeats per frequency, are the case most favourable to the theory, which predicts two levels there. The theory says what the levels would be if a neuron implemented the optimal code; these recordings do not show them.

VIII. DISCUSSION

On the evidence for five to twenty-four rates, the number of rates an optimal neuron uses in long windows is a Fisher-length quantity. The Poisson count is then the Gaussian count at the Fisher half-length $\sqrt{A}$ shortened by a wall constant, and the constant is half the extra length that a chain of levels needs next to silence. The neuron’s statistics enter only through the type of its endpoint at silence: a degenerate output at finite Fisher distance gives the Poisson wall, a regular endpoint the Gaussian one. A baseline rate turns the first type into the second, and a neuron more regular or more variable than Poisson keeps the Poisson wall and adds one finite-size term. For a rate code this means that a neuron with peak count A and baseline λ uses $N_G(\sqrt{A + \lambda} - \sqrt{\lambda} - c)$ distinct rates, that its lowest rate is silence, where it fires at the baseline rate, and that its second rate sits $(g_1/2 + \sqrt{\lambda})^2 - \lambda$ spikes per window above the baseline, 4.6 for $\lambda = 1$.

In short windows the normal approximation chose the on-off code, silence and one rate, in every case we computed, though most of those cases lie outside its accuracy range. The long-window code becomes graded past the first count change, where it soon carries much more than any on-off code, at least 24% more at $A = 10$, and at smaller peak counts when energy is scarce, where at $A = 3$ it gains less than one per cent. Near the first count change the finite-window correction thus changes which code is optimal more than it changes how much information the code carries. For neurons that are more regular than Poisson, at $A = 3$, the way the regularity is modelled matters: as spike history it leaves the capacity unchanged, as count regularity it raises it.

This work has the following limits. The wall constant is measured from the chain, and the chain law is an interpolation on $1.43 \leq g \leq 2.35$, extrapolated by 0.02 to 0.09 in g for $K = 21$ to 25 and above its range for the count changes to three and four rates; neither is derived. The count law is a conjecture tested for $K = 5$ to 24 and supported by certificates at single peak counts; it has not been evaluated at the certified 46 and 66 rates. The short-window fronts and maximizers come from a search and carry no certificate, and every short-window case with at most about 60 expected spikes per decision, including all at $n = 10$, lies past the clean validity of (3). Windows are treated as independent uses of the channel, so a GLM’s history is reset at each window. In the data, the test asked a narrow question, and a negative answer does not show that neurons ignore energy or information.

Four outcomes would change the conclusions: a certified count change at $K = 25$ outside the registered tolerance of ± 0.15 in A around $A = 412.43$, predicted on 9 September 2026 and not yet computed; a discrete level count in a spike dataset with enough trials; a count whose variance-to-mean ratio varies and that misses (6) by more than 0.02 in the half-length; and a graded short-window optimum under an exact finite-blocklength bound.

What comes next. The wall constant is measured, not derived; a derivation would make the count law a statement about every channel whose output degenerates at silence. A dataset with many trials per stimulus, counted in short windows, would decide whether neurons use discrete levels. Behind both sits a broader question that this paper does not settle: the companion paper [18] proposes that the optima of capacity, minimax estimation and the NPMLE under Gaussian noise follow one description of the same kind, a count set by a length and shortened at an edge. Whether the description extends from single neurons to populations, and from counts to spike timing, is open; if it does, the number of distinguishable responses of a neural population could be read from its noise and its range in the same way.

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APPENDIX A CERTIFICATES

For a candidate input the optimality equations, $D(x_k) = C$ at all K levels and $D'(x_k) = 0$ at the $K - 2$ interior ones, together with $\sum_k w_k = 1$, are $2K - 1$ equations in the K weights, the $K - 2$ interior positions and C . They were solved by Newton’s method at fifty digits, the solution was enclosed

TABLE IV
STATUS OF THE CLAIMS

claim	grade	evidence
exact number of rates at 35 peak counts, up to 66 rates at $A = 2000$ $N(A) = K$ on eleven intervals (170.4 in A) count law (4) with closure (5)	certified (ball arithmetic) proved under the monotonicity conjecture conjecture tested for $K = 5$ to 24; rms 0.0017; four changes predicted	Sec. III, App. A Theorem 1 Table II, Fig. 1
growth about $0.4002 A^{2/3}$, above a $\sqrt{A}$ fit to earlier numerics	through the companion paper's Gaussian count; lower bound separate, not reviewed	Sec. III
first count change at $A = 3.3679$	Shamai [1] (as quoted in [17]); recomputed	Sec. III
finite-size term (6), $\kappa = 0.42$	measured; two changes predicted	Sec. IV
baseline moves the wall; 25 published changes	tested on external data, no free parameter; six changes below $K = 5$	Sec. V
on-off loses at least 9%, 24%, 37% of capacity at $A = 6, 10, 18$ maximizer of the normal approximation is on-off	computed (Blahut–Arimoto value against $\log 2$) 185 of 185 cases (best-found search; most outside its accuracy range)	Sec. VI Sec. VI
long-window code graded under a budget below $A = 3.3679$; gain at most 0.8% at $A = 3$	graded at 42 of 83 budgets (one-sided test against all binary codes)	Fig. 2
GLM capacity equals the Poisson capacity at $A = 3$	void-probability lemma [37]; computed	Sec. VI
energy–capacity fronts; efficient rate moves up in short windows	long: global optimum; short: best found	Fig. 3
discrete levels in six datasets	registered rule: mixed in five, smooth in one; levels not found in tuned units	Table III

by a Krawczyk test with an analytic interval Jacobian [23], and $D(x) \leq C$ was verified on $[0, A]$ in three pieces: a lemma on $(0, 10^{-8}]$, cubic bounds around each level, and chord bounds with enclosed curvature on every remaining cell. The output alphabet is truncated at $A + 18\sqrt{A} + 60$ counts and the discarded tail is enclosed. The box has radius 10^{-28} or less, far below the smallest gap and the smallest weight, so the enclosed input has exactly K distinct levels with positive weights. Table V lists these thirty-two certificates.

The three later certificates use a second pipeline. The alphabet is truncated at $A + 16\sqrt{A} + 60$ counts for $A = 311.37$ and 1144.25 and at $A + 20\sqrt{A} + 60$ for $A = 2000$, and a separately proved lemma bounds the effect of the discarded tail, so the statement holds for the untruncated channel. The Krawczyk boxes have radii 10^{-22} , 10^{-26} and 10^{-30} ; near each level $D'' < 0$ is shown by a centred form, and Taylor bounds cover the stretches between levels. The smallest margins are 1.9×10^{-20} , 1.8×10^{-25} and 1.8×10^{-29} , the last about eighteen times the box radius.

APPENDIX B WHERE A NEW RATE APPEARS

Measured from the silent side in the Fisher coordinate, an inserted rate (odd K) appears at 0.62 of its gap for $K = 5$ to 21 (0.61 to 0.62 on the grid of Barletta and Dytso for $K = 5$ to 9 and 0.57 at $K = 11$, where the grid point lies furthest above the birth; 0.620 to 0.623 from forty-digit birth states for $K = 17$ to 21), and a split (even K) happens at 0.12 of the gap next to the parent rate. The Gaussian channel inserts at the centre of its central gap, $\frac{1}{2}$, and splits its central level in place; Barletta and Dytso study how its support evolves as the amplitude grows [43]. A baseline carries both fractions to the Gaussian values: the insertion position falls from 0.62 through 0.61, 0.58 and 0.56 at $\lambda = 0.01, 0.03$ and 0.05 to 0.513 and 0.507 at $\lambda = 1$ and 3, and the split position falls from 0.124 to 0.06 at $\lambda = 0.05$. The negative binomial, which keeps the Poisson wall, keeps both positions (0.624 at $\rho = 300, 0.128$ at $\rho = 3000$).

TABLE V
CERTIFIED COUNTS $N(A) = K$ (WORST CHORD SLACK IN ARB)

A	K	slack	A	K	slack
27.9452	5	5.5e-05	218.5000	17	3.4e-08
28.3620	6	2.0e-06	236.0000	17	6.2e-08
39.6120	6	2.9e-05	236.9199	18	7.5e-12
40.0280	7	1.3e-05	237.0356	18	8.3e-12
52.5555	7	2.8e-07	237.1841	18	4.9e-11
66.6505	8	7.9e-07	257.9236	18	5.5e-08
97.9375	10	1.0e-06	259.9822	19	5.0e-10
100.0000	11	1.0e-06	283.6680	20	1.4e-12
115.2578	11	1.2e-08	302.0000	20	4.0e-08
133.3550	12	1.5e-07	308.1187	21	9.8e-11
150.0000	13	1.7e-06	327.0000	21	1.9e-08
152.0000	13	3.6e-07	333.1990	22	3.3e-13
166.6681	14	1.1e-06	352.0000	22	1.4e-08
172.4467	15	7.1e-09	359.5000	23	2.8e-11
185.0000	15	5.7e-07	378.0000	23	6.7e-09
204.7761	16	3.3e-07			
214.6855	17	1.9e-09			

Later: $K = 21$ at $A = 311.37$, 46 at 1144.25, 66 at 2000.

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